BIOL4815-srk0219(a)

2025-04-26

Import and load necessary packages

```
library("DECIPHER") #MSA
```

```
## Loading required package: Biostrings
```

```
## Warning: package 'Biostrings' was built under R version 4.4.2
```

```
## Loading required package: BiocGenerics
```

```
##
## Attaching package: 'BiocGenerics'
```

```
## The following objects are masked from 'package:stats':
##
##     IQR, mad, sd, var, xtabs
```

```
## The following objects are masked from 'package:base':
##
##     anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##     colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##     get, grep, grepl, intersect, is.unsorted, lapply, Map, mapply,
##     match, mget, order, paste, pmax, pmax.int, pmin, pmin.int,
##     Position, rank, rbind, Reduce, rownames, sapply, saveRDS, setdiff,
##     table, tapply, union, unique, unsplit, which.max, which.min
```

```
## Loading required package: S4Vectors
```

```
## Loading required package: stats4
```

```
##
## Attaching package: 'S4Vectors'
```

```
## The following object is masked from 'package:utils':
##
##     findMatches
```

```
## The following objects are masked from 'package:base':
##
##     expand.grid, I, unname
```

```
## Loading required package: IRanges
```

```
## Warning: package 'IRanges' was built under R version 4.4.2
```

```
##
## Attaching package: 'IRanges'
```

```
## The following object is masked from 'package:grDevices':
##
##     windows
```

```
## Loading required package: XVector
```

```
## Loading required package: GenomeInfoDb
```

```
## Warning: package 'GenomeInfoDb' was built under R version 4.4.2
```

```
##
## Attaching package: 'Biostrings'
```

```
## The following object is masked from 'package:base':
##
##     strsplit
```

```
library("ape") #Phylogenetic trees
```

```
## Warning: package 'ape' was built under R version 4.4.3
```

```
##
## Attaching package: 'ape'
```

```
## The following object is masked from 'package:Biostrings':
##
##     complement
```

```
library("cubar") #codon usage bias
```

```
## Warning: package 'cubar' was built under R version 4.4.3
```

```
library("Biostrings") #handling and manipulating sequences
library("ggplot2") #data visualizations
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
library("ggrepel") ##data visualizations
```

```
## Warning: package 'ggrepel' was built under R version 4.4.3
```

```
library("phangorn") #phylogenetic trees
```

```
## Warning: package 'phangorn' was built under R version 4.4.3
```

Multiple Sequence Alignment

```
sequences <- readAAStringSet("C:\\Users\\sophi\\OneDrive\\Documents\\ClassAssignments\\BIOL4815
\\projectfastaproteinseq.txt")
  class(sequences)
```

```
## [1] "AAStringSet"
## attr(,"package")
## [1] "Biostrings"
```

```
sequences ##Read in all sequences in FASTA format to AAStringSet object, and verify process
```

```
## AAStringSet object of length 28:
##      width seq                                               names
##  [1]   171 MGVIEDMMKVGMRSAKAGLEATE...LLIVRAPKAEASEIEVKPMEEE  Methanopyrus kan...
##  [2]   142 MSVYEDIRRRLERLRERIVREID...SAISVRVRPNKIVEIIIPKRRQ  Hyperthermus but...
##  [3]   115 MEEFKKAMEEISKSLQKMVDDLR...HGVLIITAKKVKEEEIRIPVRG  Pyrobaculum aero...
##  [4]   166 MVRRRRWDIWDPFDLIREIQEEI...EIRIPKKNPTKKEGEGFEVKIE  Pyrococcus horik...
##  [5]   127 MEPIKKIEEGLKELYAANVGKIE...DGVLQISAIKAGEVGTAELKID  Thermoproteus uz...
##  ...   ... ...
## [24]   181 MSLIPSFIGGNRSNGFDPFSMDV...NFDLLLDISIGIYTGFNLRSRK  Lotus japonicus ...
## [25]   161 MSLIPSFFGGRRSSVFDPFSLDV...LTITVPKEELKKPDVKSIEISG  Ziziphus jujuba ...
## [26]   152 MSVYWPDYKFGDFENSVSRLFDD...ENGILEVKLPKDENAAKKISIQ  Gigaspora margar...
## [27]   154 MSIIGYGGDNFGRFESSISNMFD...QGVLEVKIPKGEAPGSKKINVQ  Acaulospora colo...
## [28]   179 MMYPSAFCCIFLFQSQKQISFNM...DGVLNLFLPKGEASKPRKITIS  Dissophora ornat...
```

```
aligned <- AlignSeqs(sequences) ##Align sequences and store as AAStringSetObject
```

```
## Determining distance matrix based on shared 5-mers:
## ================================================================================
##
## Time difference of 0.01 secs
##
## Clustering into groups by similarity:
## ================================================================================
##
## Time difference of 0.01 secs
##
## Aligning Sequences:
## ================================================================================
##
## Time difference of 0.59 secs
##
## Iteration 1 of 2:
##
## Determining distance matrix based on alignment:
## ================================================================================
##
## Time difference of 0 secs
##
## Reclustering into groups by similarity:
## ================================================================================
##
## Time difference of 0.02 secs
##
## Realigning Sequences:
## ================================================================================
##
## Time difference of 0.29 secs
##
## Iteration 2 of 2:
##
## Determining distance matrix based on alignment:
## ================================================================================
##
## Time difference of 0 secs
##
## Reclustering into groups by similarity:
## ================================================================================
##
## Time difference of 0.01 secs
##
## Realigning Sequences:
## ================================================================================
##
## Time difference of 0.2 secs
##
## Refining the alignment:
## ================================================================================
```

```
##
## Time difference of 0.22 secs
```

```
  class(aligned)
```

```
## [1] "AAStringSet"
## attr(,"package")
## [1] "Biostrings"
```

```
BrowseSeqs(aligned, title = "Cross-Domain MSA") ##Load webpage of MSA

aa_prokaryotic <- readAAStringSet("C:\\Users\\sophi\\OneDrive\\Documents\\ClassAssignments\\BIOL
4815\\aa_prokaryotic.txt")
aligned_prokaryotes <- AlignSeqs(aa_prokaryotic)
```

```
## Determining distance matrix based on shared 5-mers:
## ================================================================================
##
## Time difference of 0 secs
##
## Clustering into groups by similarity:
## ================================================================================
##
## Time difference of 0.01 secs
##
## Aligning Sequences:
## ================================================================================
##
## Time difference of 0.23 secs
##
## Iteration 1 of 2:
##
## Determining distance matrix based on alignment:
## ================================================================================
##
## Time difference of 0 secs
##
## Reclustering into groups by similarity:
## ================================================================================
##
## Time difference of 0 secs
##
## Realigning Sequences:
## ================================================================================
##
## Time difference of 0.16 secs
##
## Iteration 2 of 2:
##
## Determining distance matrix based on alignment:
## ================================================================================
##
## Time difference of 0 secs
##
## Reclustering into groups by similarity:
## ================================================================================
##
## Time difference of 0 secs
##
## Realigning Sequences:
## ================================================================================
##
## Time difference of 0.13 secs
##
## Refining the alignment:
## ================================================================================
```

```
##
## Time difference of 0.07 secs
```

```
BrowseSeqs(aligned_prokaryotes, title = "MSA of Prokaryotes")

aa_eukaryotic <- readAAStringSet("C:\\Users\\sophi\\OneDrive\\Documents\\ClassAssignments\\BIOL4
815\\aa_eukaryotic.txt")
aligned_eukaryotes <- AlignSeqs(aa_eukaryotic)
```

```
## Determining distance matrix based on shared 5-mers:
## ================================================================================
##
## Time difference of 0 secs
##
## Clustering into groups by similarity:
## ================================================================================
##
## Time difference of 0 secs
##
## Aligning Sequences:
## ================================================================================
##
## Time difference of 0.2 secs
##
## Iteration 1 of 2:
##
## Determining distance matrix based on alignment:
## ================================================================================
##
## Time difference of 0 secs
##
## Reclustering into groups by similarity:
## ================================================================================
##
## Time difference of 0.01 secs
##
## Realigning Sequences:
## ================================================================================
##
## Time difference of 0.13 secs
##
## Iteration 2 of 2:
##
## Determining distance matrix based on alignment:
## ================================================================================
##
## Time difference of 0 secs
##
## Reclustering into groups by similarity:
## ================================================================================
##
## Time difference of 0.01 secs
##
## Realigning Sequences:
## ================================================================================
##
## Time difference of 0.01 secs
##
## Refining the alignment:
## ================================================================================
```

```
##
## Time difference of 0.08 secs
```

```
BrowseSeqs(aligned_eukaryotes, title = "MSA of Eukaryotes")
```

Distance Matrix and Principal Component Analysis

```
dists <- DistanceMatrix(aligned, type = "matrix", correction = "Poisson") #DECIPHER
```

```
## ================================================================================
##
## Time difference of 0 secs
```

```
View(dists) ##Create distance matrix with Poisson correction method
class(dists)
```

```
## [1] "matrix" "array"
```

```
max_dist <- max(dists, na.rm = TRUE)
print(max_dist)
```

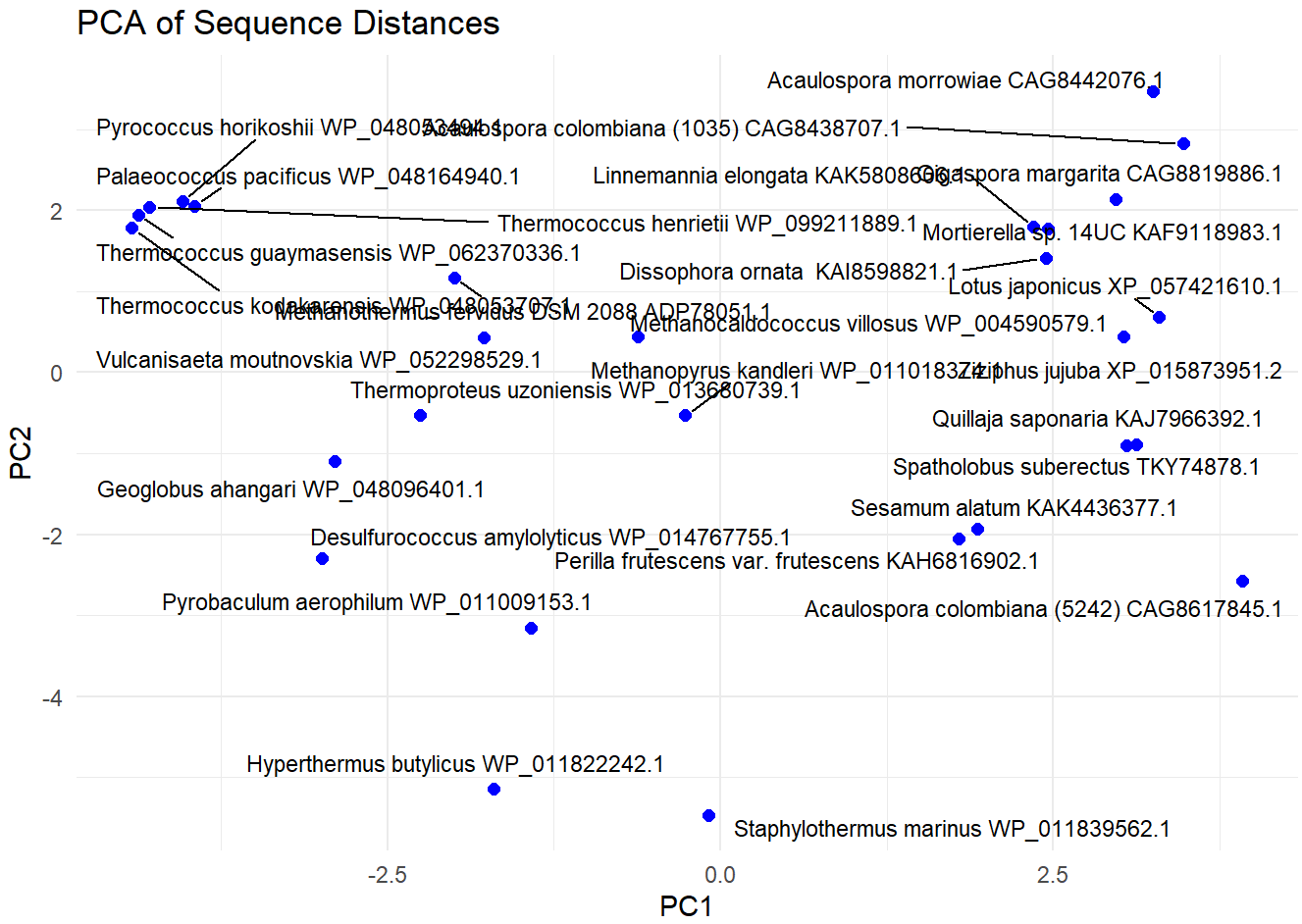
```
## [1] 2.369525
```

```
##Perform Principal Component Analysis - base R
pca_result <- prcomp(dists, scale. = TRUE)

# Create a data frame from PCA results
pca_df <- data.frame(
  PC1 = pca_result$x[,1],
  PC2 = pca_result$x[,2],
  Label = rownames(dists)
)
class(pca_df)
```

```
## [1] "data.frame"
```

```
# Plot using ggplot2
ggplot(pca_df, aes(x = PC1, y = PC2, label = Label)) +
  geom_point(color = "blue", size = 2) +
  geom_text_repel(size = 3, box.padding = 0.4, point.padding = 0.3, max.overlaps = 50) +
  theme_minimal() +
  labs(title = "PCA of Sequence Distances",
       x = "PC1", y = "PC2")
```




```
summary(pca_result)
```

```
## Importance of components:
##                           PC1    PC2    PC3     PC4    PC5     PC6    PC7
## Standard deviation     2.9447 2.3157 1.8855 1.59600 1.2432 1.12846 1.0504
## Proportion of Variance 0.3097 0.1915 0.1270 0.09097 0.0552 0.04548 0.0394
## Cumulative Proportion  0.3097 0.5012 0.6282 0.71915 0.7743 0.81982 0.8592
##                           PC8     PC9    PC10    PC11    PC12   PC13    PC14
## Standard deviation     0.91059 0.86390 0.81583 0.69743 0.57127 0.5674 0.47260
## Proportion of Variance 0.02961 0.02665 0.02377 0.01737 0.01166 0.0115 0.00798
## Cumulative Proportion  0.88884 0.91549 0.93926 0.95664 0.96829 0.9798 0.98777
##                           PC15   PC16    PC17    PC18    PC19    PC20    PC21
## Standard deviation     0.35747 0.2946 0.23271 0.14870 0.13134 0.11127 0.08846
## Proportion of Variance 0.00456 0.0031 0.00193 0.00079 0.00062 0.00044 0.00028
## Cumulative Proportion  0.99233 0.9954 0.99736 0.99815 0.99877 0.99921 0.99949
##                           PC22    PC23    PC24    PC25    PC26    PC27
## Standard deviation     0.07621 0.07075 0.04419 0.02562 0.02425 0.01599
## Proportion of Variance 0.00021 0.00018 0.00007 0.00002 0.00002 0.00001
## Cumulative Proportion  0.99970 0.99988 0.99995 0.99997 0.99999 1.00000
##                           PC28
## Standard deviation     1.152e-15
## Proportion of Variance 0.000e+00
## Cumulative Proportion  1.000e+00
```

```
head(pca_result$x)
```

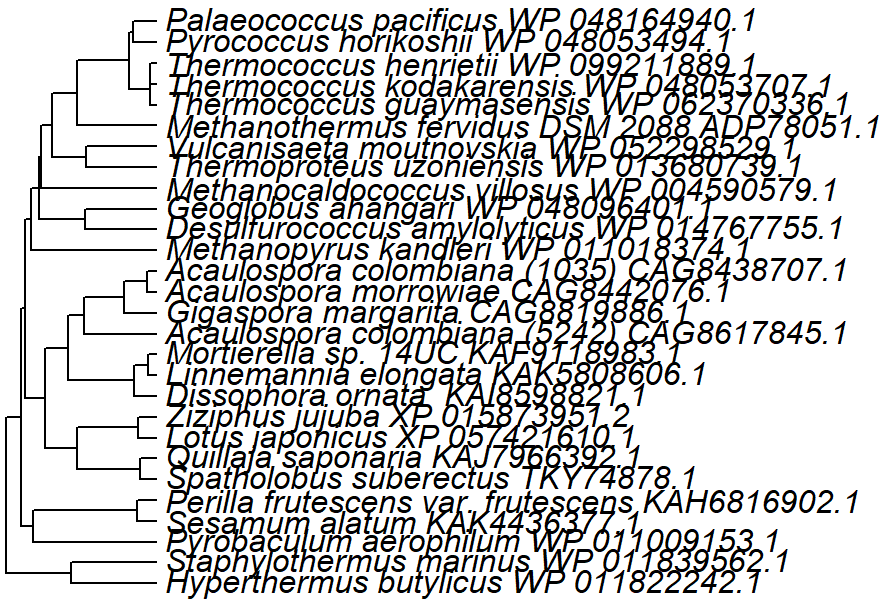
```
##                                                PC1        PC2        PC3
##  Methanopyrus kandleri WP_011018374.1     -0.2623767 -0.5291875 -1.3921920
##  Hyperthermus butylicus WP_011822242.1    -1.6990505 -5.1472407  2.3249306
##  Pyrobaculum aerophilum WP_011009153.1    -1.4197798 -3.1522090 -1.0696965
##  Pyrococcus horikoshii WP_048053494.1     -4.0394272  2.1106007 -0.5862387
##  Thermoproteus uzoniensis WP_013680739.1  -2.2524658 -0.5218030 -1.0244677
##  Thermococcus kodakarensis WP_048053707.1 -4.4194567  1.7850972 -0.4675691
##                                                PC4         PC5         PC6
##  Methanopyrus kandleri WP_011018374.1      0.8887923 -1.24020006  0.11309065
##  Hyperthermus butylicus WP_011822242.1    -2.5556843 -0.13692310  1.09398295
##  Pyrobaculum aerophilum WP_011009153.1     3.0475835 -0.39446067 -0.63559369
##  Pyrococcus horikoshii WP_048053494.1     -0.8549226  0.37837670  0.26470336
##  Thermoproteus uzoniensis WP_013680739.1   1.7699046 -2.72117244  0.44063887
##  Thermococcus kodakarensis WP_048053707.1 -1.3425354  0.01886532  0.01629932
##                                                PC7        PC8          PC9
##  Methanopyrus kandleri WP_011018374.1     -3.3950170  1.6492249 -1.240759840
##  Hyperthermus butylicus WP_011822242.1    -1.0833611  0.1620262 -0.642986817
##  Pyrobaculum aerophilum WP_011009153.1     0.8900576  2.3740555  0.369461601
##  Pyrococcus horikoshii WP_048053494.1      0.2190486  0.2544933  0.206096465
##  Thermoproteus uzoniensis WP_013680739.1  -1.4196948 -0.6628094  1.107813664
##  Thermococcus kodakarensis WP_048053707.1  0.7782345  0.5594184 -0.007281751
##                                                PC10       PC11        PC12
##  Methanopyrus kandleri WP_011018374.1      1.03296726 -0.8508642  0.12165048
##  Hyperthermus butylicus WP_011822242.1    -0.07266488 -0.3771493  1.17881671
##  Pyrobaculum aerophilum WP_011009153.1    -1.34598498  1.5336876 -0.11366199
##  Pyrococcus horikoshii WP_048053494.1      0.50190991  0.1167000 -0.12622003
##  Thermoproteus uzoniensis WP_013680739.1  -0.37994673  0.8283318  0.44191956
##  Thermococcus kodakarensis WP_048053707.1  0.34635339  0.1771404 -0.07905127
##                                                 PC13        PC14        PC15
##  Methanopyrus kandleri WP_011018374.1      0.3663034 -0.93007837 -0.04654447
##  Hyperthermus butylicus WP_011822242.1    -0.3648096  0.80317604 -0.13815251
##  Pyrobaculum aerophilum WP_011009153.1     0.7857758 -0.10434696 -0.19083189
##  Pyrococcus horikoshii WP_048053494.1     -0.2304604 -0.26862554  0.22130854
##  Thermoproteus uzoniensis WP_013680739.1  -0.4264299  1.06221848  0.29639833
##  Thermococcus kodakarensis WP_048053707.1 -0.3028087  0.04996513 -0.10635700
##                                                PC16         PC17        PC18
##  Methanopyrus kandleri WP_011018374.1      0.2066809 -0.008461296  0.02090197
##  Hyperthermus butylicus WP_011822242.1    -0.5858409  0.219474125 -0.10779197
##  Pyrobaculum aerophilum WP_011009153.1    -0.3030413  0.181898331 -0.02302455
##  Pyrococcus horikoshii WP_048053494.1      0.2284104  0.216151123 -0.30778903
##  Thermoproteus uzoniensis WP_013680739.1   0.5606048 -0.356655264  0.04739788
##  Thermococcus kodakarensis WP_048053707.1 -0.2518889 -0.081507459  0.13471163
##                                                 PC19          PC20
##  Methanopyrus kandleri WP_011018374.1      0.041573495  1.365705e-02
##  Hyperthermus butylicus WP_011822242.1    -0.081062751  2.307182e-02
##  Pyrobaculum aerophilum WP_011009153.1    -0.041809993  1.643872e-03
##  Pyrococcus horikoshii WP_048053494.1     -0.098581098 -3.422154e-01
##  Thermoproteus uzoniensis WP_013680739.1  -0.006382798 -6.185121e-05
##  Thermococcus kodakarensis WP_048053707.1  0.104319215 -9.421922e-04
##                                                 PC21          PC22
##  Methanopyrus kandleri WP_011018374.1     -0.001124588  0.0009425489
##  Hyperthermus butylicus WP_011822242.1     0.011208649 -0.0047097353
```

```
##  Pyrobaculum aerophilum WP_011009153.1    -0.014487514 -0.0245337162
##  Pyrococcus horikoshii WP_048053494.1     -0.073005573  0.0045055527
##  Thermoproteus uzoniensis WP_013680739.1  -0.003537163  0.0331694066
##  Thermococcus kodakarensis WP_048053707.1 -0.003961901  0.0262248656
##                                                 PC23         PC24         PC25
##  Methanopyrus kandleri WP_011018374.1     -0.02724342 -0.011559138 -0.003613282
##  Hyperthermus butylicus WP_011822242.1     0.03060853  0.006479827  0.001476580
##  Pyrobaculum aerophilum WP_011009153.1     0.01554674  0.005283788  0.002031537
##  Pyrococcus horikoshii WP_048053494.1      0.14564181  0.028179741 -0.007548897
##  Thermoproteus uzoniensis WP_013680739.1   0.02102794 -0.001221867 -0.002548386
##  Thermococcus kodakarensis WP_048053707.1 -0.05646150 -0.012689898 -0.071278499
##                                                   PC26          PC27
##  Methanopyrus kandleri WP_011018374.1     -0.0027604399  0.0008031377
##  Hyperthermus butylicus WP_011822242.1     0.0041535776  0.0017864416
##  Pyrobaculum aerophilum WP_011009153.1     0.0034761756  0.0007913137
##  Pyrococcus horikoshii WP_048053494.1      0.0008348808 -0.0050169371
##  Thermoproteus uzoniensis WP_013680739.1  -0.0044301759  0.0009044871
##  Thermococcus kodakarensis WP_048053707.1 -0.0696236931  0.0209559295
##                                                   PC28
##  Methanopyrus kandleri WP_011018374.1      6.245005e-16
##  Hyperthermus butylicus WP_011822242.1    -2.498002e-16
##  Pyrobaculum aerophilum WP_011009153.1     1.047773e-15
##  Pyrococcus horikoshii WP_048053494.1      9.783840e-16
##  Thermoproteus uzoniensis WP_013680739.1   5.724587e-16
##  Thermococcus kodakarensis WP_048053707.1 -3.469447e-18
```

##Phylogenetic Tree

```
set.seed(42)

tree_upgma <- upgma(dists)
plot(tree_upgma, main = "UPGMA Phylogenetic Tree")
```




```
bootstrap_trees <- boot.phylo(tree_upgma, dists, FUN = function(xx) upgma(dist.aa(xx)), B = 100,
trees = TRUE)
```

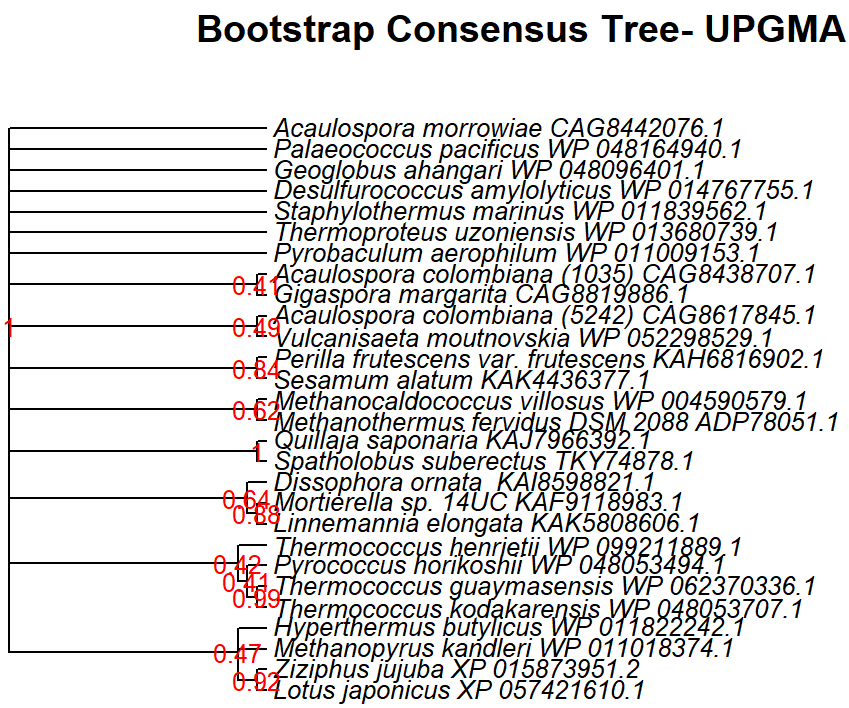
```
## Running bootstraps:       100 / 100
## Calculating bootstrap values... done.
```

```
print(bootstrap_trees) ##Run bootstrapping of the tree w given distance matrix using 100 iterati
ons and UPGMA formula
```

```
## $BP
##  [1] 100  NA  NA  NA  NA  NA  NA  NA  84  NA  NA  NA  NA  NA  NA  NA  64  NA  NA
## [20] 100  92  NA  NA  38  NA  88  NA
##
## $trees
## 100 phylogenetic trees
```

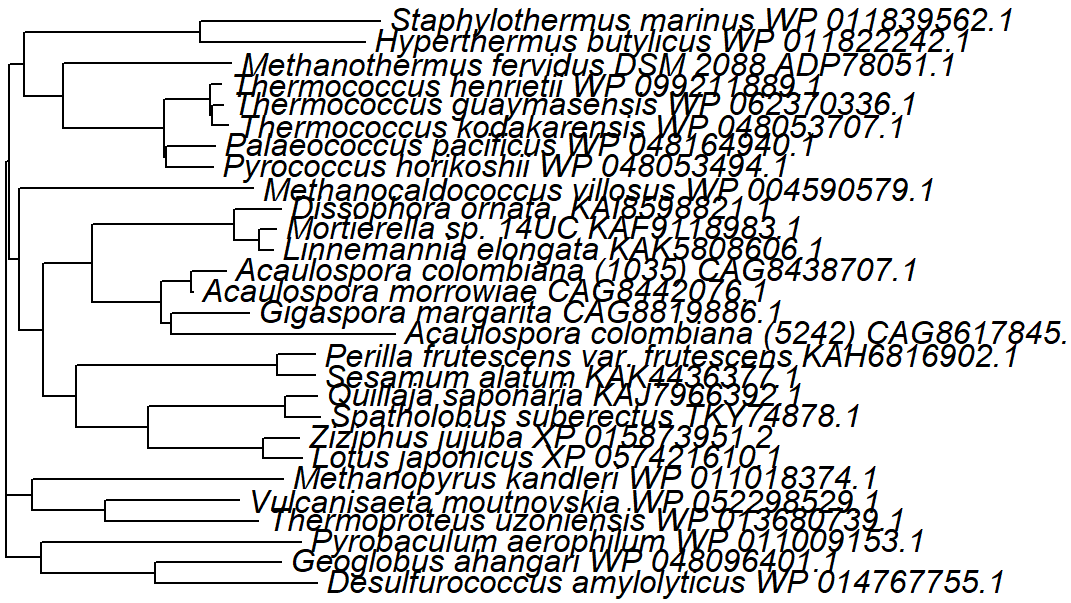
```
consensus_tree <- consensus(bootstrap_trees$trees, p = 0.39, check.labels = TRUE, rooted = TRUE)
consensus_tree <- addConfidences(consensus_tree, bootstrap_trees$trees)

# Add to plot
plot(consensus_tree, main="Bootstrap Consensus Tree- UPGMA", type="phylogram", cex = 0.8, cex.la
b = 1.5, show.tip.label = TRUE)
nodelabels(consensus_tree$node.label, frame = "none", cex = 0.8, cex.lab = 1.5, col = "red")
```




```
tree_nj <- nj(dists)
plot(tree_nj, main = "Neighbor Joining Phylogenetic Tree")
```

Neighbor Joining Phylogenetic Tree




```
nj_bootstrap_trees <- boot.phylo(tree_nj, dists, FUN = function(xx) nj(dist.aa(xx)), B = 100, tr
ees = TRUE)
```

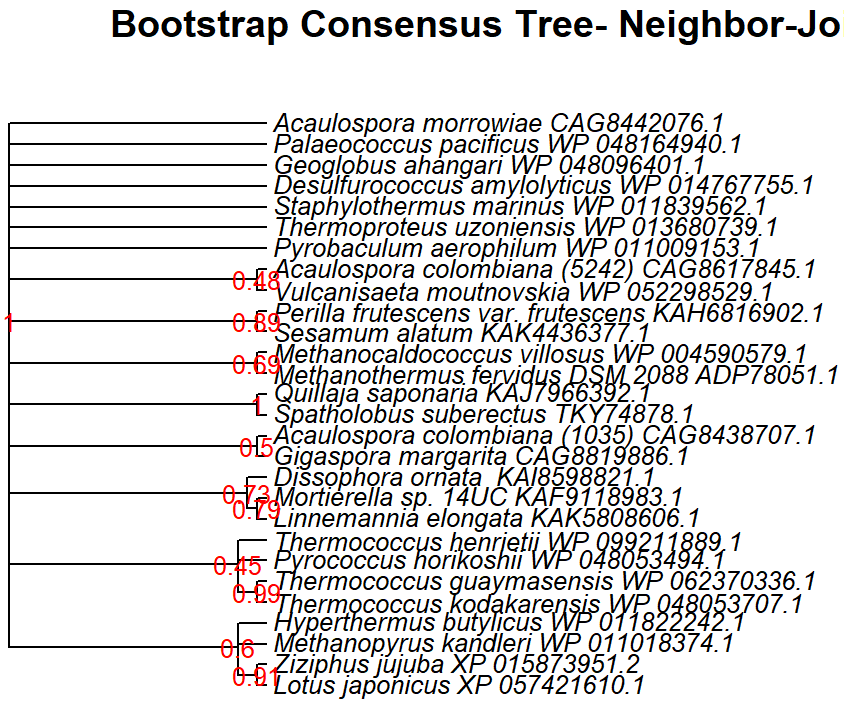
```
## Running bootstraps:       100 / 100
## Calculating bootstrap values... done.
```

```
print(nj_bootstrap_trees) ##Run bootstrapping of the tree w given distance matrix using 100 iter
ations and UPGMA formula
```

```
## $BP
##  [1]  NA   0   0   0   0   0   2   0   0   0   0   0   1   0   0   0   2   3  41
## [20]  99   0  73  79  91 100  89
##
## $trees
## 100 phylogenetic trees
```

```
nj_consensus_tree <- consensus(nj_bootstrap_trees$trees, p = 0.45, check.labels = TRUE, rooted = FALSE)
nj_consensus_tree <- addConfidences(nj_consensus_tree, nj_bootstrap_trees$trees)

# Add to plot
plot(nj_consensus_tree, main="Bootstrap Consensus Tree- Neighbor-Joining", type="phylogram", cex = 0.8, cex.lab = 1.5, show.tip.label = TRUE)
nodelabels(nj_consensus_tree$node.label, frame = "none", cex = 0.8, cex.lab = 1.5, col = "red")
```



##GC Content Analysis

```
dna_sequences <- readDNAStringSet("C:\\Users\\sophi\\OneDrive\\Documents\\ClassAssignments\\BIOL4815\\projectfastadnaseq.txt")
  print(dna_sequences)
```

```
## DNAStringSet object of length 28:
##      width seq                                               names
##  [1]   516 GTGGGCGTGATCGAGGACATGAT...GAAACCGATGGAGGAAGAGTGA  Methanopyrus kan...
##  [2]   429 ATGTCGGTCTACGAGGATATTAG...CATACCGAAGAGGAGGCAGTAG  Hyperthermus but...
##  [3]   387 ATGGAACCAATTAAGAGAATTGA...AGCTGAGCTGAAAATTGACTAG  Pyrobaculum aero...
##  [4]   501 ATGGTGAGGAGGAGAAGGTGGGA...ATTTGAAGTTAAGATCGAGTGA  Pyrococcus horik...
##  [5]   528 ATGTCGGTGTTCGACGAGTTCGA...ATTCGAAATAAAGGTCGAATAA  Thermoproteus uz...
##  ...   ... ...
## [24]  1155 ATTGTGTTTCCATTTCTGAAATT...TTTGGATTCTTTATTTGTGTTT  Lotus japonicus ...
## [25]   814 AAGCCACCATCTTCTTCTTCCCT...TATATTATATATTGGTTTTTTT  Ziziphus jujuba ...
## [26]   545 ATGTCAGTATACTGGCCTGATTA...TAAGAAAATTTCAATTCAATAA  Gigaspora margar...
## [27]   553 ATGTCGATTATTGGTTACGGTGG...CAAGAAGATTAACGTTCAGTAA  Acaulospora colo...
## [28]   677 GGAGAAATGTATATAAATCTTCG...AAAGAGATTGCAATGGAAAACA  Dissophora ornat...
```

```
gc_content <- letterFrequency(dna_sequences, letters = c("G", "C"), as.prob=TRUE)
gc_total <- rowSums(gc_content)
print(gc_total)
```

```
##  [1] 0.5988372 0.5268065 0.4935401 0.4431138 0.5530303 0.5843621 0.3634752
##  [8] 0.4125230 0.4559585 0.3081761 0.5480226 0.5734990 0.5720165 0.3109620
## [15] 0.4383838 0.5371550 0.4249714 0.3870023 0.3577982 0.4307562 0.6165803
## [22] 0.5424028 0.5696203 0.3844156 0.4299754 0.3192661 0.4141049 0.4711965
```

```
summary(gc_total)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.3082  0.4061  0.4495  0.4667  0.5493  0.6166
```

```
max_gc <- which.max(gc_total)
cat("Highest GC content is in ", names(dna_sequences)[max_gc])
```

```
## Highest GC content is in   Perilla frutescens var. frutescens QFCC02000001.1:c65044347-650437
69
```

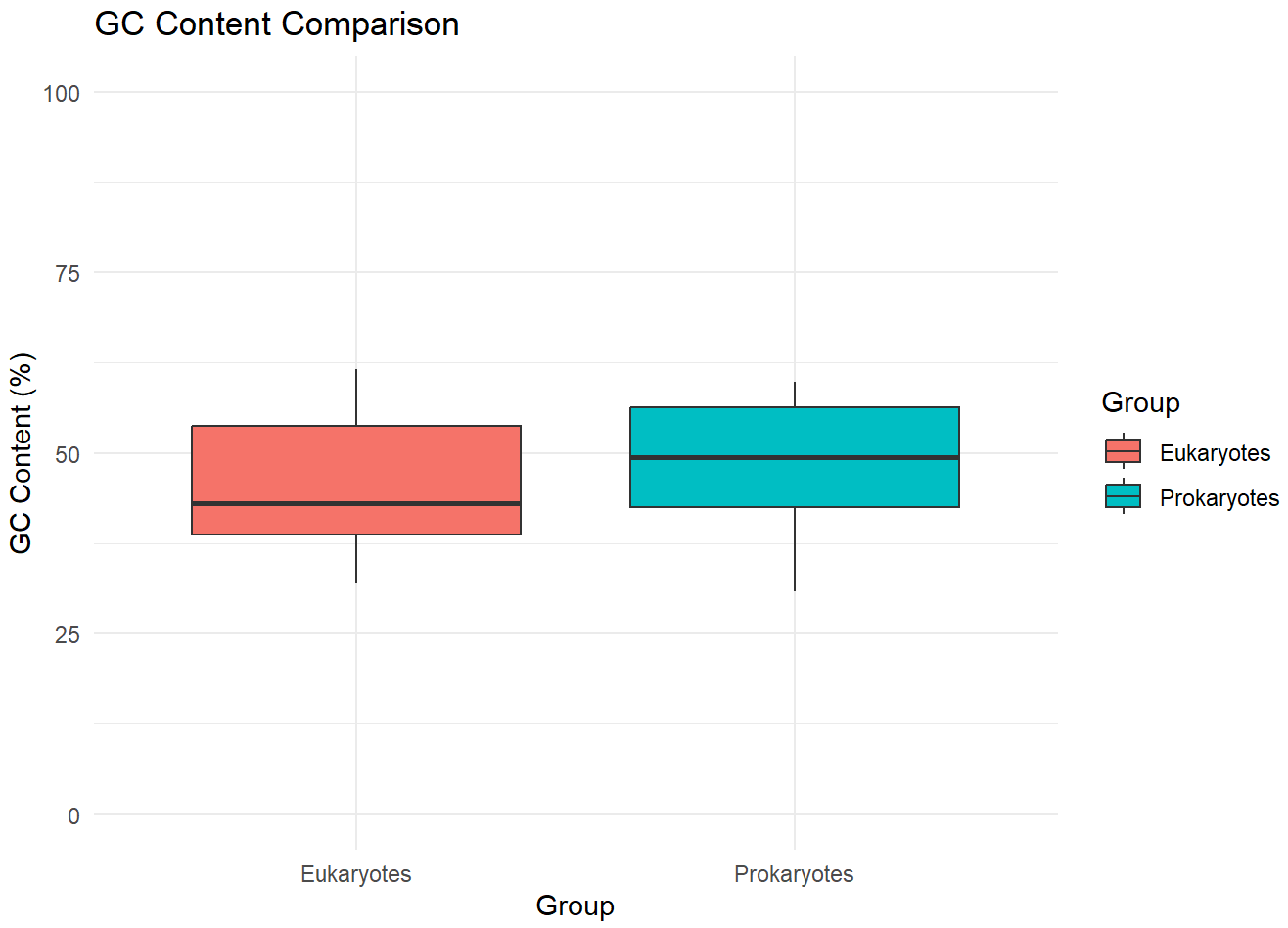
```
dna_prokaryotic <- readDNAStringSet("C:\\Users\\sophi\\OneDrive\\Documents\\ClassAssignments\\BIOL4815\\dna_prokaryotic.txt")
dna_eukaryotic <- readDNAStringSet("C:\\Users\\sophi\\OneDrive\\Documents\\ClassAssignments\\BIOL4815\\dna_eukaryotic.txt")

gc_prok <- rowSums(letterFrequency(dna_prokaryotic, c("G", "C"), as.prob=TRUE)) * 100
gc_euk <- rowSums(letterFrequency(dna_eukaryotic, c("G", "C"), as.prob=TRUE)) * 100

gc_data <- data.frame(
  GC_Content = c(gc_prok, gc_euk),
  Group = c(rep("Prokaryotes", length(gc_prok)), rep("Eukaryotes", length(gc_euk)))
)

ggplot(gc_data, aes(x=Group, y=GC_Content, fill=Group)) +
  geom_boxplot() +
  scale_y_continuous(limits = c(0, 100))+  # Y-axis from 0 to 1

  theme_minimal() +
  labs(title="GC Content Comparison", y="GC Content (%)")
```




```
t.test(gc_prok, gc_euk)
```

```
##
##  Welch Two Sample t-test
##
## data:  gc_prok and gc_euk
## t = 0.73931, df = 25.928, p-value = 0.4664
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -4.654066  9.881260
## sample estimates:
## mean of x mean of y
##  47.88471  45.27111
```

Codon Usage Bias

```
cds_prokaryotic <- readDNAStringSet("C:\\Users\\sophi\\OneDrive\\Documents\\ClassAssignments\\BI
OL4815\\cds_prokaryotic.txt")
seq_prokaryotic <- as.character(cds_prokaryotic)

cds_eukaryotic <-  readDNAStringSet("C:\\Users\\sophi\\OneDrive\\Documents\\ClassAssignments\\BI
OL4815\\cds_eukaryotic.txt")
seq_eukaryotic <- as.character(cds_eukaryotic)

# Function to clean sequences
clean_seqs <- function(seqs) {
  seqs <- toupper(seqs)
  seqs <- seqs[nchar(seqs) %% 3 == 0]  # Keep sequences with length divisible by 3
  return(seqs)
}

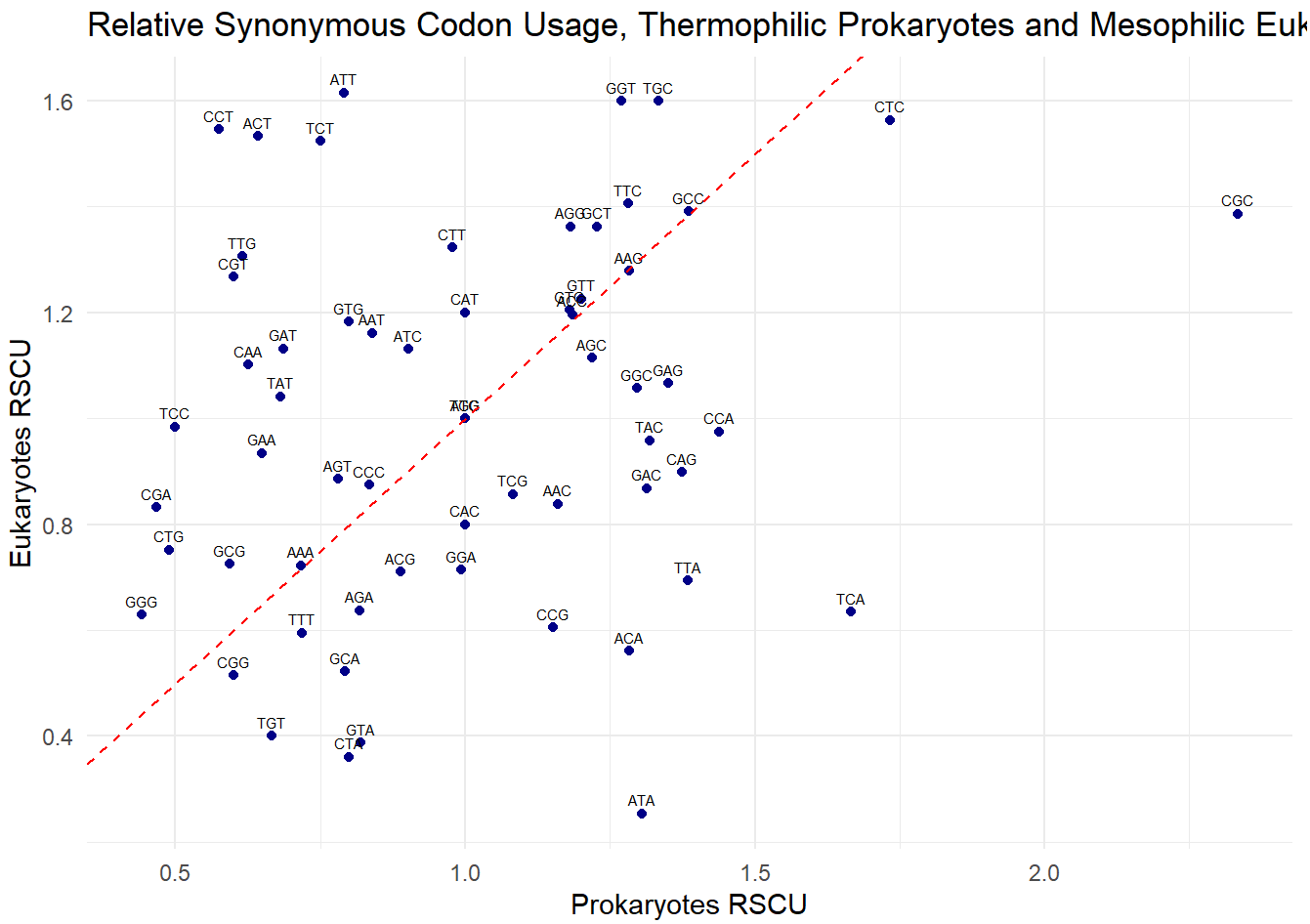
# Clean sequences
seqs_prok_clean <- clean_seqs(seq_prokaryotic)
seqs_euk_clean  <- clean_seqs(seq_eukaryotic)

codon_prok <- count_codons(seqs_prok_clean)
codon_euk <- count_codons(seqs_euk_clean)

rscu_prok <- est_rscu(codon_prok)
rscu_euk  <- est_rscu(codon_euk)

# Merge RSCU data frames
rscu_combined <- merge(rscu_prok, rscu_euk, by = "codon", suffixes = c("_prok", "_euk"))

ggplot(rscu_combined, aes(x = rscu_prok, y = rscu_euk, label = codon)) +
  geom_point(color = "darkblue") +
  geom_text(size = 2, vjust = -0.7) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "red") +
  labs(title = "Relative Synonymous Codon Usage, Thermophilic Prokaryotes and Mesophilic Eukaryo
tes",
       x = "Prokaryotes RSCU",
       y = "Eukaryotes RSCU") +
  theme_minimal()
```


Relative Synonymous Codon Usage, Thermophilic Prokaryotes and Mesophilic Euk
Eukaryotes RSCU
Prokaryotes RSCU
1.6
1.2
0.8
0.4
0.5
1.0
1.5
2.0
ATT
GGT
TGC
CTC
CCT
ACT
TCT
TTC
GCC
CGC
AGC
GCT
CTT
TTG
CGT
AAG
GTT
CAT
GTG
AAT
ATC
GAT
CAA
AGC
GGC
GAG
TAT
TCC
TAC
CCA
GAA
AGT
CCC
CAG
GAC
TCG
AAC
CGA
CAC
CTG
GCG
AAA
ACG
GGA
TTA
GGG
AGA
TCA
TTT
CCG
ACA
CGG
GCA
TGT
GTA
CTA
ATA